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SUPERSEDING
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MILITARY SPECIFICATIONS

PRIMER MARK 102 MOD 1

1. SCOPE

1.1 This specification governs the manufacture, assembly, loading, and preparation for delivery of Primer Mk 102 Mod 1 and the methods of inspection and tests upon which acceptance will be based.

2. APPLICABLE DOCUMENTS

2.1 The following standards, drawings, and publications of the issue in effect on date of invitation for bids, form a part of this specification:

STANDARDS

MIL-STD-105	--	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	--	Marking for Shipment and Storage.

DRAWINGS

BUREAU OF ORDNANCE

LD 275909	--	Primer Mk 102 Mod 1,
LD 166538	--	Test Set (Detonator) Mk 136 Mod 0,
LD 168793	--	Detonator Container Mk 3 Mod 0,
LD 168794	--	Detonator Container Mk 2 Mod 0,
553491	--	Test Ball,
and all documents		listed on the above LDs and drawings.

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PUBLICATIONS

BUREAU OF ORDNANCE

OP 400 -- General Instructions for the Design, Manufacture, and Inspection of Naval Ordnance Equipment.

OD 5823 -- Sensitivity Test of Primers and Detonators using Test Set Mk 135 Mod 0 (Primer) and Test Set Mk 136 Mod 0 (Detonator).

(Copies of standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. -- The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 71-78 -- Transportation, Interstate Commerce Commission, Explosives and Other Dangerous Articles.

(The Interstate Commerce Commission regulations are now a part of the Code of Federal Regulations (1949 Edition-Revised 1950) available from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Orders for the above publication should cite "49 CFR 71-78 (Rev. 1950).")

3. REQUIREMENTS

3.1 General. -- Primer Mk 102 Mod 1 covered by this specification shall be manufactured in accordance with Bureau of Ordnance List of Drawings LD No. 275909, and all documents listed thereon.

3.1.1 The requirements and procedures established in OP 400 shall be adhered to wherever applicable.

3.2 Primer mixture. -- The mixture of ingredients specified by the applicable drawing, shall be a thoroughly mixed uniform powder.

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3.3 Moisture content. -- The moisture content of the priming mixture at the time of loading shall be less than 0.30 percent as determined by test procedure specified in 4.5.1.

3.4 Loading. -- The priming mixture shall be loaded by ram force as specified on the applicable drawings. Loading and assembly operations shall be carried out under conditions which will keep the moisture content of the explosive charges in the primers less than 0.30 percent by weight.

3.5 Primer sensitivity. -- The primers shall be capable of functioning by the low impact energy of a firing pin. The sensitivity of functioning shall be ascertained by means of Test Set (Detonator) Mk 136 Mod 0 with a two ounce hardened steel ball to drive the firing pin. The testing procedure shall be as specified in 4.5.2. The sensitivity values shall be used for data card record and process control. Deviations in sensitivity values shall not directly cause the rejection of primers, but may be reason for checking materials and processes for compliance with drawing and specification requirements.

3.6 Primer functioning. -- The primers shall function at a 3.5 inch drop height with a 2 ounce drop ball using Test Set (Detonator) Mk 136 Mod 0. Testing and acceptance shall be as specified in 4.5.3.

3.7 Primer output. -- The explosion of the primer in the Test Set Mk 136 Mod 0 shall produce an indentation in the lead disc equal to or greater than 0.035 inch. Testing and acceptance shall be as specified in 4.5.4.

4. QUALITY ASSURANCE PROVISIONS AND TEST REQUIREMENTS

4.1 Sampling. -- Unless otherwise specified, and when applicable, the sampling plans and procedures used by the Government Inspector in the determination of the acceptability of products submitted by a supplier for Government Inspection will be in accordance with the provisions of Standard MIL-STD-105.

4.1.1 Lot. -- As applied to the Government Inspection of units of product, the term "Lot" shall mean "Inspection Lot", i.e., a collection of units of the product used as a

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basis for Government Inspection. Unless otherwise specified, the number of units of product in "Inspection Lots" shall be as determined by the Government Inspector, and may differ from the quantity designated in the contract or order as a lot for production, shipment, or other purpose.

4.2 Preproduction sample. -- Before entering into quantity production, a preproduction sample of six hundred (600) primers Mk 102 Mod 1 shall be delivered by the Contractor for test to an agency designated by the Procuring Agency. These primers shall be manufactured by the same procedures and processes and at the same location proposed by the Contractor for the execution of the contract. When a preproduction sample is received which proves to be satisfactory for the purpose intended, the Contractor will be notified and will be authorized by the Procuring Agency to proceed with the submission of the product for acceptance. Accepted samples will become the property of the Procuring Agency and will be included in the number called for in the contract schedule.

4.2.1 Preproduction sample for a subsequent contract. -- The necessity for a preproduction sample will be determined by the Procuring Agency when production under a new contract by the same Contractor at the same location follows the manufacture of satisfactory material as covered by this specification.

4.3 Production lot. -- For procurement purposes each lot of loaded primers shall consist of 10,000 loaded primers plus those required for all test purposes. Unless otherwise stated in the contract or order, or requisition all units for test purposes shall be provided at the expense of the contractor. The production lot should not be confused with the lot for inspection purposes defined in 4.1.1.

4.4 Inspection. -- The general inspection procedures shall be in accordance with "General Instructions for the Design, Manufacture, and Inspection of Naval Ordnance Equipment", OP 400; Government Inspection shall be in accordance with the provisions of General Procedures for Government Inspection and Contractor's Inspection shall be in accordance with Inspection by Contractor, as contained in OP 400.

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4.5 Tests.

4.5.1 Moisture content test for priming mixture. -- Conduct at least one test for each lot of primers and such additional tests as the Inspector considers warranted by variations in the process and materials. Collect an approximate two gram sample of the material being loaded in a tared, stoppered container at the site of and immediately prior to consolidating the charge in the primer. Weigh the container; open and expose the sample for drying at 50 ± 2 degrees C and at least 28 inches (Hg) vacuum until constant weight is reached. Cool the container in a desiccator; stopper and weigh. All weighings shall be accurate to ± 0.2 milligram. The presence of moisture in excess of 0.30 percent shall cause suspension of the charge loading until permanent corrective measures satisfactory to the Inspector have been completed. Primers which have been loaded with the defective charges shall be rejected.

4.5.2 Primer sensitivity test. -- Select 50 primers from each lot (no lot shall exceed 10,000 primers) for the Bruceton (staircase) Sensitivity Test employing 1.0 inch increments. Use Test Set (Detonator) Mk 136 Mod 0, Bureau of Ordnance List of Drawings LD No. 166538, with Test Ball, Bureau of Ordnance Drawing No. 553491, piece 3. Test set operation and test procedures are described in OD 5823. Record, on the Ammunition Data Card specified in 5.4, the 50 percent firing height ($\bar{h}$) and the standard deviation (s) calculated from the test data.

4.5.3 Primer functioning test. -- Samples shall be selected and tested from each lot in accordance with the Single Sampling Plan, Inspection Level II, AQL 0.25 percent defective as given in Standard MIL-STD-105. The initiation of the primer shall be accomplished by the use of Test Set (Detonator) Mk 136 Mod 0 equipped with the test ball specified in 4.5.2. The two ounce ball shall be dropped from a height of 3.5 inches to initiate the primer. Primers which do not explode on the first penetration of the firing pin shall be classed as defective. Acceptance of the lot shall be in accordance with the sampling plan given above. The indented lead discs shall be retained and used to fill the sample requirements specified in 4.5.4.

4.5.4 Primer output test. -- Samples shall be selected and tested in accordance with the single sampling plan, Inspection Level II, AQL 1.0 percent defective as given in Standard MIL-STD-105. The indented lead discs retained from 4.5.3 shall be used to fill the sample requirements.

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A primer which produces an indentation in the lead disc less than 0.035 inch shall be classed as defective. Acceptance of the lot shall be in accordance with the sampling plan given above. The procedure for measuring the lead disc indentations shall be as specified in 4.5.4.1.

4.5.4.1 Test procedure. -- Each lead disc shall be measured for the average depth of indentation to the nearest thousandth of an inch. The following method of obtaining average depth readings shall be used. Mount a dial indicator or comparator vertically on a stand so that the tip of the indicator is perpendicular to the surface plate. The indicator shaft shall be 0.100 inch in diameter and shall have a 60 degree conical tip with a 0.025 inch radius on the point. Place the lead disc, indented side up, upon a ring type holder so that the periphery of the bottom of the lead disc is parallel to the surface plate. Lower the indicator tip to the bottom of the indentation of the lead disc and set the dial reading at zero. The lead disc measurement shall be the average of the readings of four equidistant points near the periphery of the upper surface of the lead disc.

Note: Prior to measurement of the depth of indentation, aluminum pieces of the primer cup embedded in the lead disc which interfere with the measurement shall be removed by the use of a sharp pointed instrument. Care shall be taken in removal of the embedded aluminum particles to prevent gouging of the disc.

5. PREPARATION FOR DELIVERY

5.1 Packaging. -- Primers Mk 102 Mod 0 shall be unit packaged in accordance with Bureau of Ordnance List of Drawings No. 168793.

5.2 Packing. -- Fifty (50) unit containers packaged as specified in 5.1 shall be packed in accordance with Bureau of Ordnance List of Drawings LD No. 168794.

5.3 Marking. -- In addition to any special marking required by the contract or order, all containers shall be marked in accordance with Standard MIL-STD-129, and Code of Federal Regulations 49 CFR 71-78.

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5.4 Data cards. -- Data cards, five by eight inches, printed as shown below shall be filled out for each lot of primers. Two of these cards shall accompany the shipping ticket, three shall be forwarded to the Government and five shall be furnished to the Inspector.

AMMUNITION DATA CARD

Mk No. _____ Spec. No. _____

Dwg. No. _____ Date (Mo. & Yr.) _____

Loaded by _____ Req. No. _____

Contract No. _____ No. in Lot _____

No. in Box _____ h _____

Lot No. _____ s _____

MATERIAL	MANUFACTURER	LOT NO.	DATE (MO. & YR.)
Cup			
Sealing disc			
Basis lead Styphnate			
Tetracene			
Antimony Sulfide			
Barium Nitrate			
Lead Azide			
Priming mixture			

Remarks: (These are to include any changes in materials or manufacturing processes prescribed by this specification which the Inspector feels might affect the uniformity of the lot.)

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6. NOTES

6.1 Intended use. -- Primer Mk 102 Mod 1 is used as the initiating element in fuze explosive trains. This primer is used in fuzes Mk 19, Mk 20, Mk 21, Mk 28, Mk 31, Mk 36, Mk 39, Mk 48, Mk 64, Mk 161 and Mk 166.

Patent notice. -- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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